DroneOps Intelligence

Mission planning · live monitoring incident analytics · advisory AI

Design → database → packages → states → demo workflows

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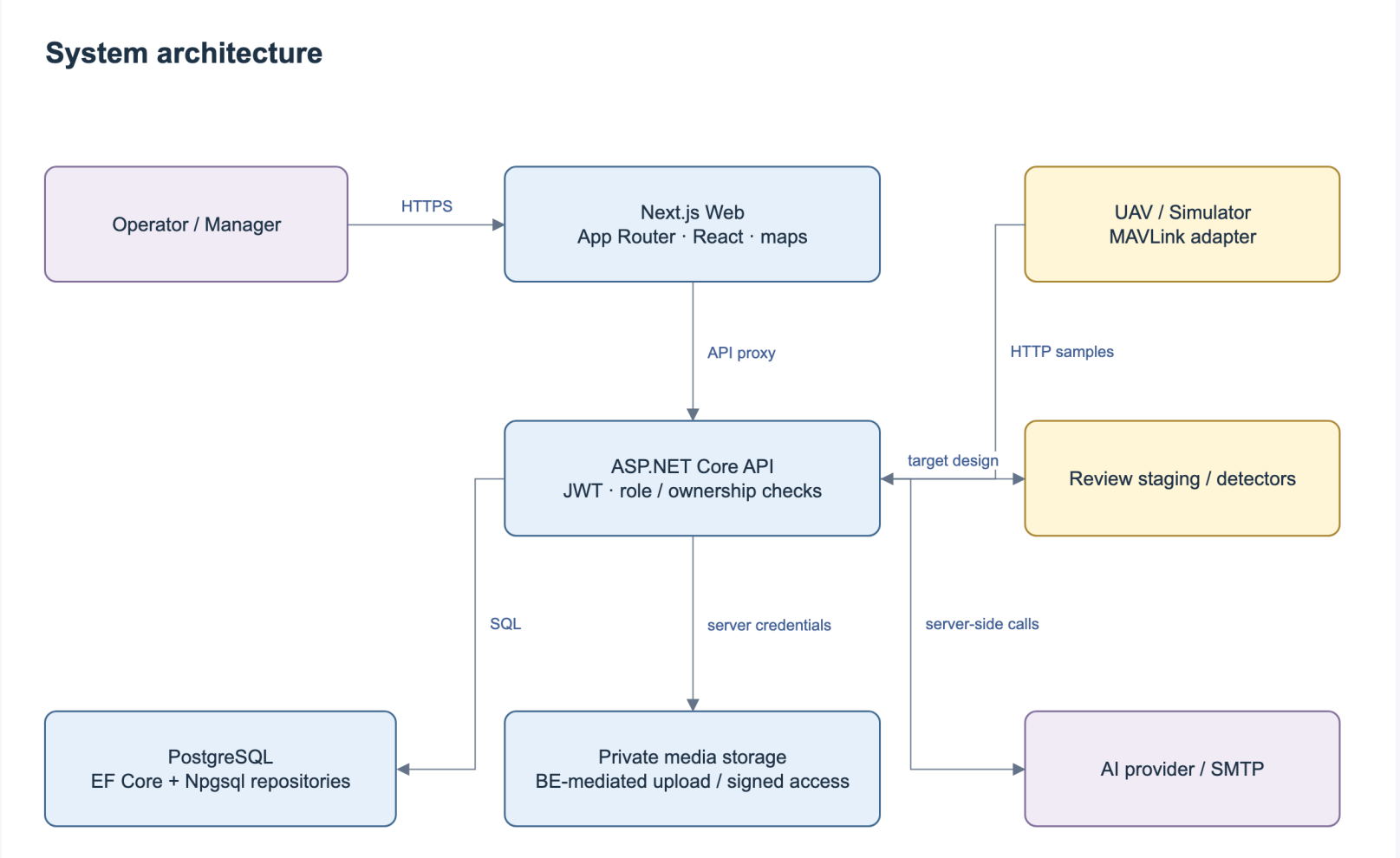
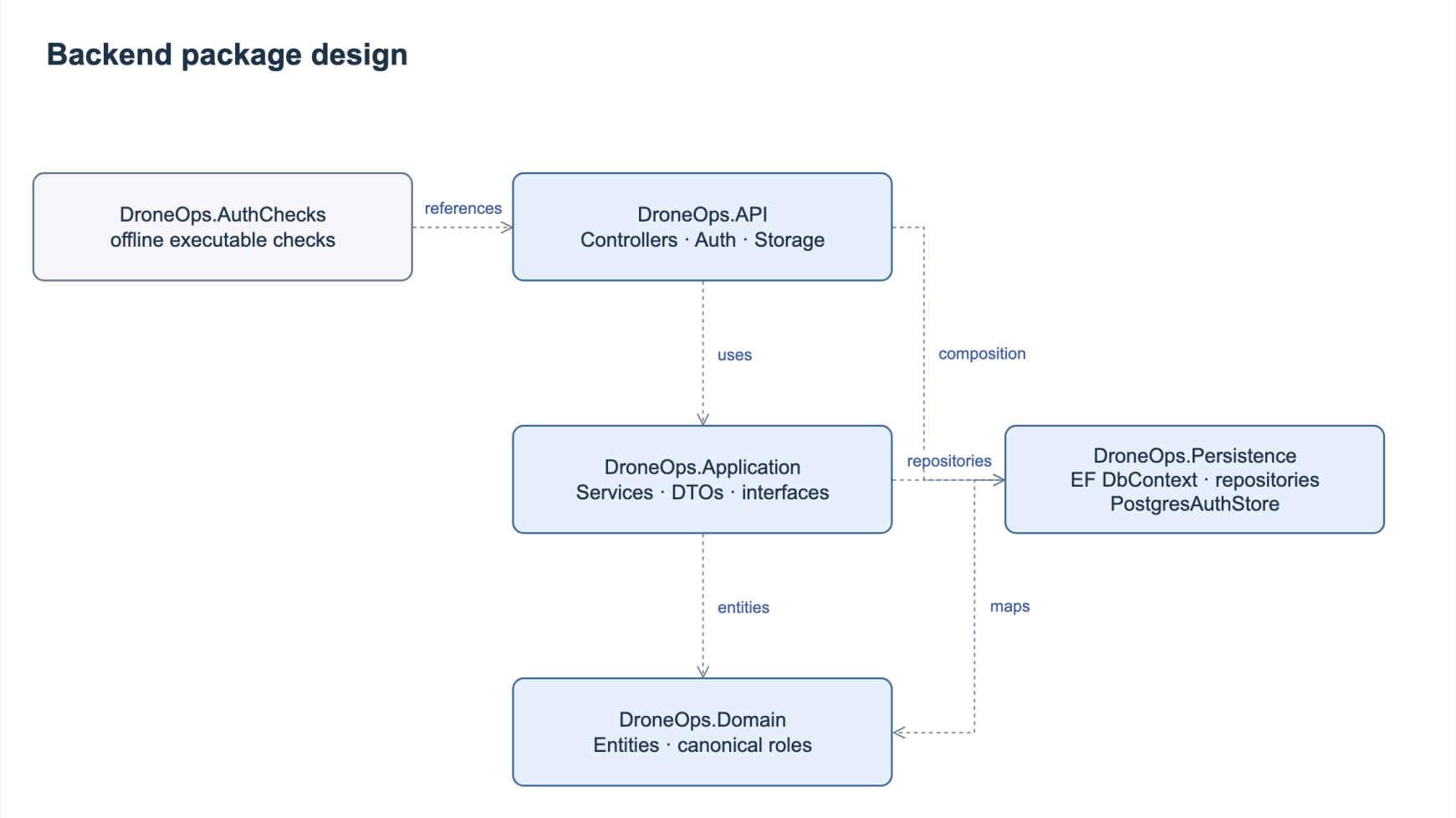
System design and integration boundaries



Package design follows the actual repository



Web packages

Portal pages

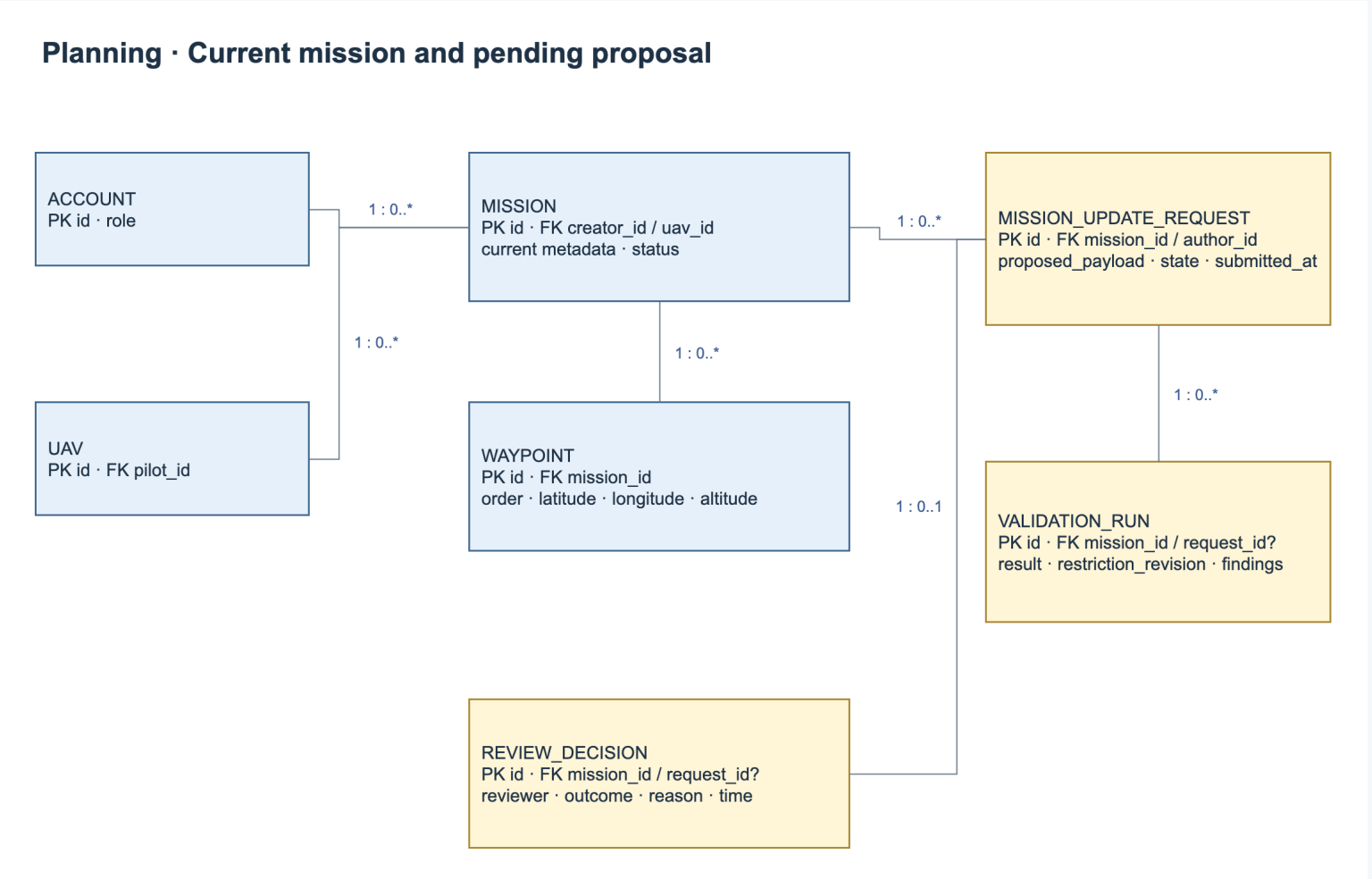
↓

Shared clients

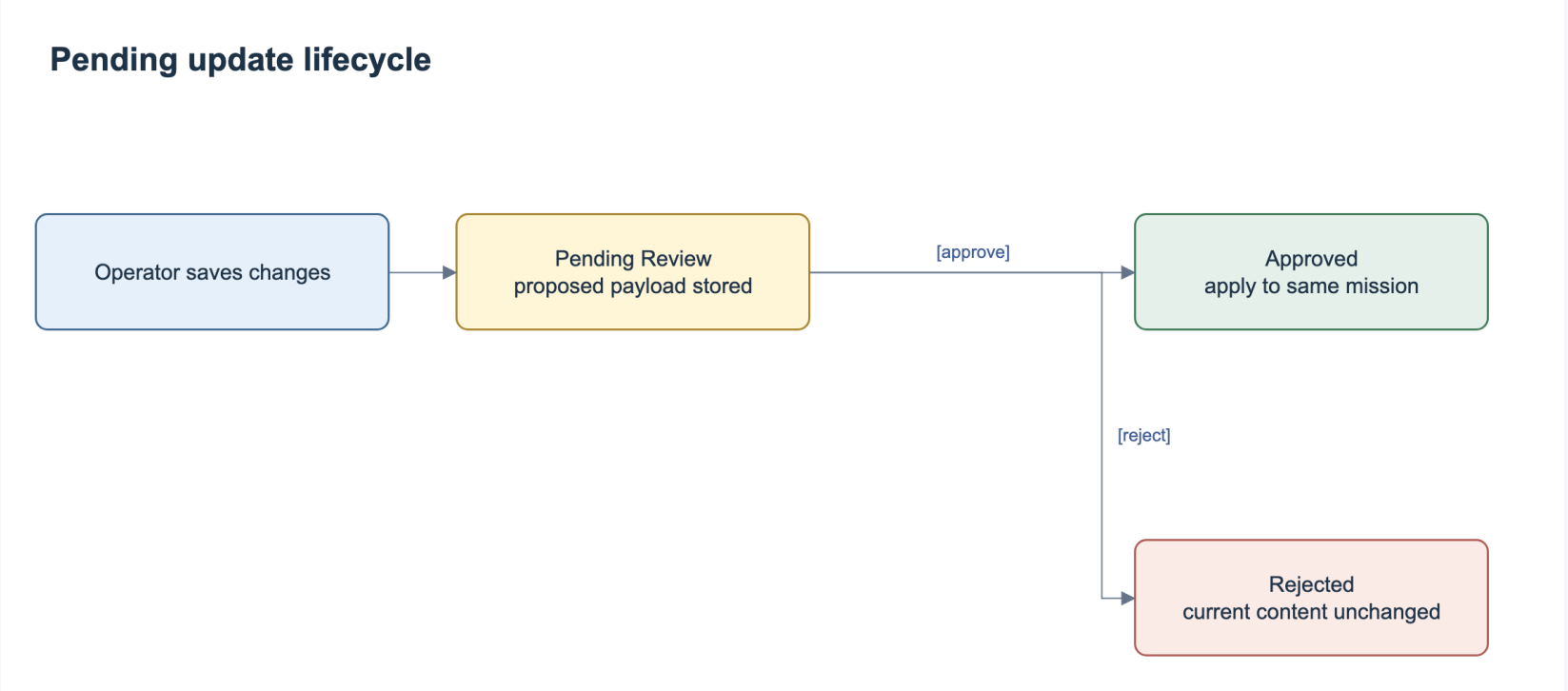
↓

API proxy

Database: current content and proposed updates

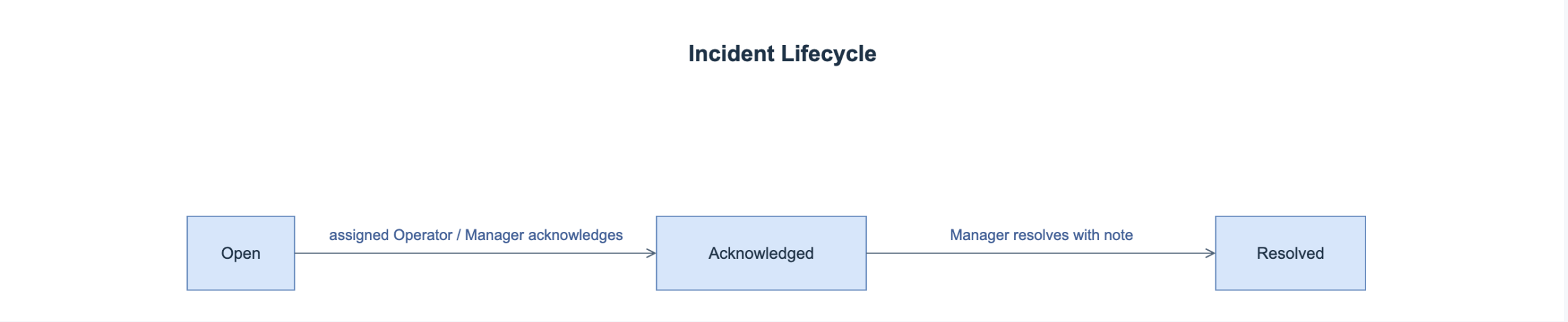


Current mission and update request have separate states



Approved + rejected update → original content and approval remain

Incident states preserve source evidence



Frozen evidence + actor/time/reason + durable alerts

Target actions still require API and FE integration.

Workflow A–C: review every mission update

1 · Save update

Store proposed content; send Manager review.

2 · Review

Compare old/new content and current validation.

3 · Reject

Keep old content and prior Approved status.

4 · Approve

Apply all accepted changes to the same mission.

Workflow D: telemetry → monitoring → incidents

Receive and persist

Authenticated normalized samples; source/receive times; deduplicate.

Monitor and reconnect

Latest snapshot + SSE; freshness, unavailable and offline states.

Detect and record

Unit-aware rules; independent loss timer; frozen evidence.

Handle and notify

Acknowledge → Manager resolve; durable alert and timeline.

History and AI retain a traceable evidence chain

Post-flight history

Known telemetry fixture → report metrics, incidents and coverage.

Advisory AI

Authorized context → provider result or explicit unavailable state.

Open scope item: registered Top-k similarity evidence versus current provider-only integration.

Test design targets the critical invariants

Mission invariants

Reject preserves current data; approve applies atomically; stale evidence blocks.

Realtime reliability

Out-of-order/replay, null metrics, no-packet timeout and reconnect.

40 planned integration/system cases + 10 unit designs

All Pending — capture tagged build, fixtures, logs and actual results.

CI: .NET build / analyzers / offline checks; FE lint / types / tests / build.

Workflow demonstration sequence

1. Create a mission and submit it for pre-flight validation.
2. Reject an update and verify the approved content is preserved.
3. Approve an update and verify the current mission changes.
4. Monitor telemetry and inspect incident evidence.
5. Review the post-flight report and AI assessment.

Decisions and next actions for Review 2

Agree the mission update and review rules.

Validate geometry, altitude references and boundary handling.

Complete gateway, monitoring and incident integration.

Execute test cases and record measurable results.

Review database migrations and deployment configuration.